Reg. No. | | | | | | | | | | |

[12ME 51/13ME 51]

FIFTH SEMESTER B.E/B.TECH EXAMINATION, DECEMBER-2015

PRECISION ENGINEERING

Time:3 Hours **Maximum Marks:70**

Instructions:

Answer any **FIVE** full Questions.

1. a. Define metrology. What are the objectives of metrology? **04**

   b. Describe with neat sketches. **06**

      i. Imperials standard yard

      ii. international prototype meter

   c Define the terms. **04**

      i. Accuracy

      ii. resolution, related to measuring instruments

2. a. Explain unilateral & bilateral tolerances. **04**

   b. Determine the tolerance on the hole& shaft for a fit designated by $50H_7\ g_6$; diameter step 30 to 50 mm; fundamental deviation for shaft $=-2.5\ D0^{0.34}$ in $-2.5D^{0.34}$ microns; fundamental deviator for 'H' hole =0, IT6 =10i; and IT7=16i, $i=0.45\ 3\sqrt{D}+0.001D$ in micron. **06**

   c Briefly explain. **04**

      i. Go- gauges

      ii. No-Go gauges

3. a. What are comparators? Mention any four characteristics of comparators. **04**

   b. With a neat sketch explain the principle of sine bar. **05**

   c Explain the working of LVDT with neat sketch. **05**

4. a. Explain 3-wire method of measuring effective diameter of screw thread. **08**

   b. Explain the principle of interferometry write a note on optical flat. **06**

5. a. Define the following terms with reference to measurement system. **06**

      i. calibration

      ii. sensitivity

      iii. Hysteresis

      iv. Threshold value

   b. Define error and explain the classification of errors. **08**

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6.	**a.**	With a neat sketch describe the pirani gauge used for pressure measurement.	**08**
	b.	Explain with a suitable diagram, prony brake Dynamometer.	**06**
7.	**a.**	Sketch and explain the working principle of optical pyrometer.	**08**
	b.	What is a thermocouple? State the laws of thermocouple with sketches.	**06**
8.	**a.**	With a neat sketch explain length bar measuring machine. Mention its applications.	**08**
	b.	Write a note on electronic inspection.	**06**